

Abhay Raj Gurram

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Summary

Machine Learning and AI engineering graduate student experienced in Python, PyTorch, data analysis, and machine learning systems development. Skilled in building ML pipelines, performing exploratory data analysis, and implementing AI-driven solutions using modern frameworks including LangChain and Databricks. Strong ability to collaborate across teams, analyze engineering and business data, and deploy scalable ML systems.

Education

Master of Science — Business Analytics and Artificial Intelligence, Stevens Institute of Technology (Expected 2026)

Technical Skills

Programming	Python, SQL
Machine Learning	PyTorch, Scikit-Learn, XGBoost, Regression, Classification, Neural Networks
AI & LLM Tools	LangChain, LangGraph, Prompt Engineering, Agentic AI Workflows
Data Science	Pandas, NumPy, SciPy, Feature Engineering, Data Mining, Exploratory Data Analysis
Visualization	Matplotlib, Plotly, Seaborn, Streamlit
Platforms	Databricks, Jupyter Notebook, VS Code
Development Tools	Git, Docker, FastAPI, REST APIs
Databases	Snowflake, PostgreSQL

Machine Learning Projects

AI Infrastructure Monitoring & Anomaly Detection System

Developed a machine learning system for detecting anomalies in infrastructure telemetry data using Python and XGBoost. Built feature engineering pipelines to process operational datasets and trained predictive models to identify abnormal behavior. Implemented FastAPI inference services and created dashboards to visualize anomaly detection results.

LLM Document Intelligence System

Built a document intelligence system using LangChain and Python to analyze enterprise knowledge repositories. Implemented semantic retrieval pipelines, document indexing, and NLP-based classification to extract insights from engineering and business documents.

Predictive Demand Forecasting System

Developed machine learning models to forecast operational demand using time-series datasets. Performed exploratory data analysis, engineered predictive features, evaluated model performance, and built visualization dashboards to communicate forecasting insights.

Data Analytics & AI Experience

Conducted exploratory data analysis using Python, Pandas, and NumPy to identify patterns in large datasets. Applied machine learning techniques including classification and regression to solve predictive problems. Designed data pipelines, performed feature engineering, and created visual dashboards to present analytical insights.

Soft Skills

Problem Solving | Cross-Functional Collaboration | Technical Documentation | Communication | Time Management | Project Ownership